DISSERTATION PROPOSAL PRESENTATION

CareerPilot AI

An Explainable AI-Driven Career Readiness and Personalized Skill Development Platform

Python | FastAPI | React + TypeScript | PostgreSQL | Explainable Recommendations



Presented by [Student Name] · [BITS ID]

Supervisor [Supervisor Name], [Designation]

Organisation [Organisation / Mentor Name]

BITS Pilani · CSIW ZG628T Dissertation · Academic Session 2026–2027

OVERVIEW

Presentation Outline

What we will cover in the next 10–15 minutes

01 Problem and motivation

02 Proposed solution

03 Objectives and scope

04 Architecture and workflow

05 Core methods

06 Database design

07 Evaluation plan

08 Timeline and conclusion

THE PROBLEM

Why Students Struggle With Career Preparation

The gap between choosing a career and preparing for it

- Unclear skill expectations for a chosen role
- Too much disconnected learning material
- Hard to recognise one's own weaknesses
- Uncertainty about what to learn first
- Limited weekly study time
- Generic roadmaps ignore individual results

PROBLEM STATEMENT

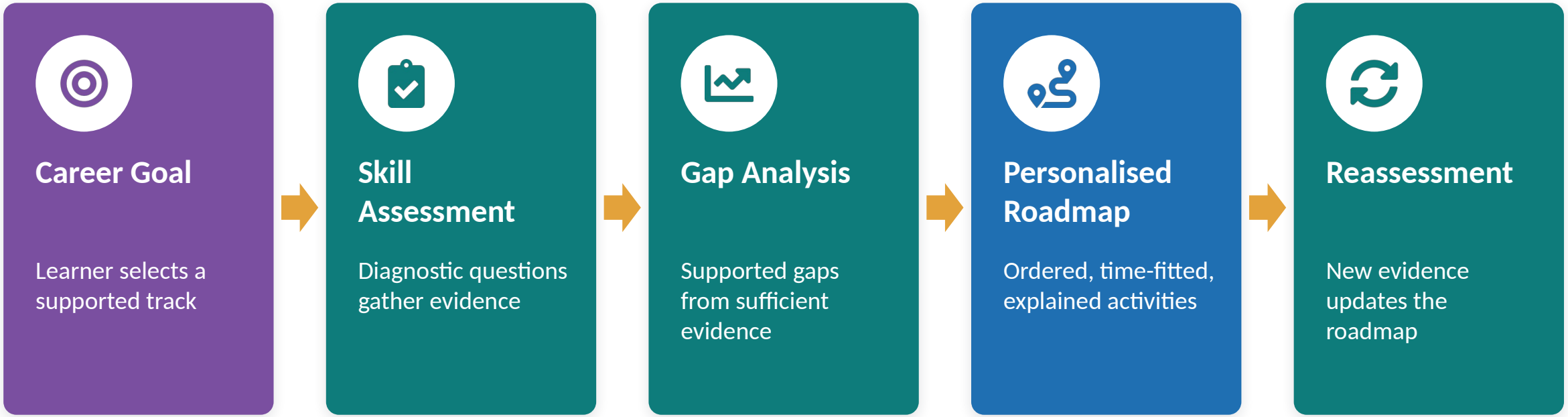
How can a web application identify evidence-supported skill gaps and generate explainable, personalised learning roadmaps for a selected software career track?

Why it matters
Learners differ in existing skills, prerequisites and available time — one roadmap cannot fit everyone.

OUR PROPOSED SOLUTION

CareerPilot AI in One Picture

An assessment-driven, explainable learning planner



In one sentence: CareerPilot AI uses a learner's assessment evidence, skill prerequisites and weekly study time to recommend an explained, personalised learning roadmap.

OBJECTIVES

What We Propose to Build

Initial career track: Junior Python Backend Developer

Assessment

- Career-track selection
- Curated role–skill database
- Diagnostic assessment
- Skill evidence and gap analysis

Recommendation

- Prerequisite-aware recommendation
- Weekly time-budget planning
- Explainable roadmap generation

Learning management

- Secure authentication
- Progress tracking
- Reassessment support
- Admin content management

RESEARCH CORE

the explainable assessment-to-roadmap recommendation mechanism.

SCOPE

What Is In — and Out — of the First Release

Final scope to be confirmed with our supervisor

In scope

- One track: Junior Python Backend Developer
- Curated skill groups (SD-01 to SD-07 to start)
- Reviewed diagnostic questions
- Free, curated learning resources
- Assessment and roadmap workflows
- Progress tracking and reassessment
- Administrator interface

Out of scope / future work

- Employment prediction
- Placement guarantees
- Multiple career tracks
- Paid LLM APIs
- Running learner-submitted code
- Job-board integration

Synopsis scope: ~20–30 skills and 100–150 reviewed items. We may start smaller and grow, subject to supervisor approval.

SYSTEM ARCHITECTURE

How React, FastAPI, Python and PostgreSQL Work Together

Proposed layered design

Student / Administrator — Browser access, authenticated by role

↓

React + TypeScript — Assessment screens, skill profile, roadmap dashboard

↓

FastAPI backend — APIs, validation, authentication, trusted server-side scoring

↓

Python recommendation engine — Evidence analysis, prerequisite graph, matching, time-budget scheduling

PostgreSQL

Users, skills, questions, attempts, estimates, resources and versioned roadmaps

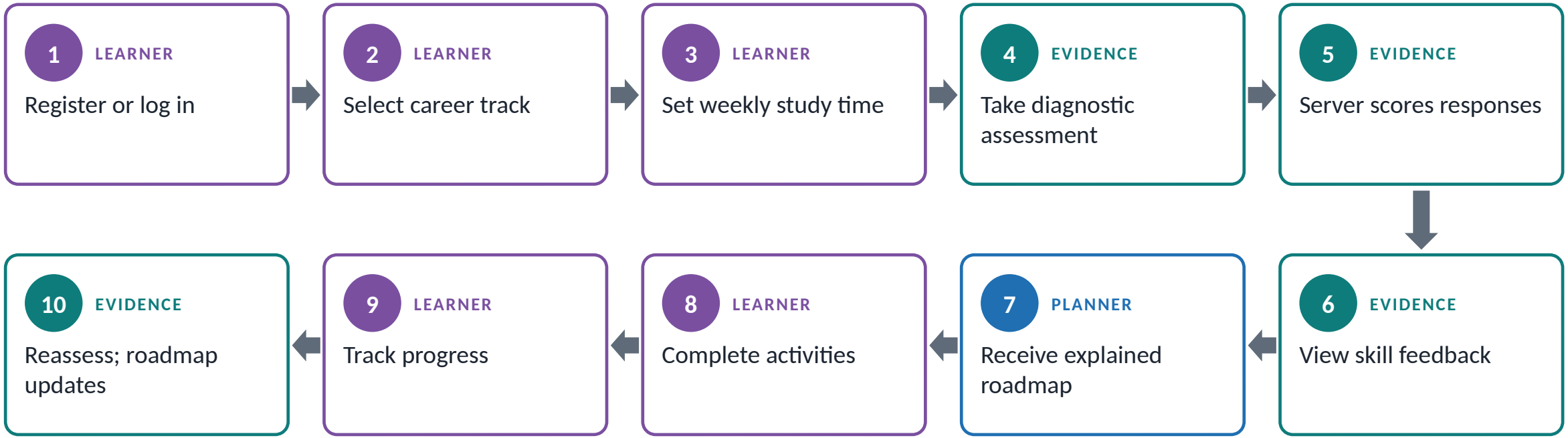
Security

Role-based access; answer keys never sent to the browser early

END-TO-END WORKFLOW

How a Learner Uses CareerPilot AI

Proposed learner journey



Important: completing an activity is recorded as progress — only an eligible reassessment creates new skill evidence.

CORE METHOD 1

Diagnostic Assessment and Evidence States

Measuring what the learner can currently demonstrate

HOW IT WORKS

- Each question is mapped to a skill (e.g. SD-01 Programming Fundamentals)
- Scoring happens on the server: correct = 1, incorrect = 0, unanswered = missing
- Score = correct eligible answers ÷ answers required by the coverage rule
- Missing evidence is never treated as zero skill

Evidence states (illustrative examples)

State	Example	System action
NOT_ASSESSED	No SD-01 answers	Offer assessment
INSUFFICIENT	1 of 2 required answered	Ask for more evidence
DIAGNOSTIC_AVAILABLE	2 of 2 answered, 1 correct	Score = 0.50

0.50 means one of two items was correct — not "50% proficient" and not a validated measure of competence.

CORE METHOD 2

Gap Analysis and Prerequisite-Aware Planning

Rule-based, evidence-driven and constraint-aware

PLANNING RULES

- Gap = max(0, target − score), only when evidence is sufficient
- Prerequisite graph (a DAG) decides the learning order
- Only published, reviewed resources are recommended
- Weekly time budget is never exceeded (e.g. 180 min/week)

PLANNER LOGIC (pseudocode)

```
for skill in prioritised(supported_gaps(evidence)):
    for res in published_resources(skill):
        if prerequisites_ok(res, plan):
            week = first_week_with_room(res.minutes)
            plan.add(res, week,
                     reasons=collect_reasons())
```

Example: 90 + 60 + 75 = 225 min does not fit a 180-minute week, so the planner moves an activity to week 2.

CORE METHOD 3

Explainable Recommendations

Every recommendation keeps the reasons that produced it

WHY IT IS EXPLAINABLE

- The planner stores reason codes with each decision
- Learner-facing text is generated from those stored codes
- No post-hoc explanations invented after the fact
- Earlier roadmaps and evidence stay in version history

Recommended (illustrative): LR-03 Algorithm Tracing
Week 2 · about 75 minutes · skill SD-02

Stored reasons
ASSESSED_GAP · RESOURCE_SKILL_MATCH · PREREQUISITE_FIRST · TIME_BUDGET_FIT

Explanation shown to the learner
"Your diagnostic evidence shows a learning need in algorithm reasoning. This activity is mapped to that skill and fits your weekly time."

DATABASE DESIGN

What PostgreSQL Will Store

Conceptual entities — final schema designed during implementation

USER	CAREER_TRACK	SKILL	SKILL_PREREQUISITE	ASSESSMENT_VERSION
role	name	SD-01 … SD-07	skill → requires	published config

QUESTION_VERSION	ASSESSMENT_ATTEMPT	QUESTION_RESPONSE	SKILL_ESTIMATE	LEARNING_RESOURCE
skill mapping	user · version	attempt · answer	evidence state · score	skill · minutes · status

ROADMAP_VERSION	ROADMAP_ACTIVITY	RECOMMENDATION_REASON	ACTIVITY_PROGRESS	ROADMAP
refs skill estimates	resource · week	reason code	state · timestamps	user · track

KEY RELATIONSHIPS

- A career track requires many skills
- Questions provide evidence for mapped skills
- Attempts contain responses; estimates reference eligible evidence
- Roadmap versions reference the skill estimates they used
- Activities reference published resources and store reasons
- Progress never edits skill estimates

EVALUATION PLAN

How We Will Evaluate the System

Planned methodology — no experiments have been run yet

WHAT WE WILL MEASURE

- Skill coverage
- Prerequisite compliance
- Weekly time-budget compliance
- Resource relevance
- Recommendation reproducibility
- Explanation consistency

ALSO TESTED

Authentication, role-based access, server-side scoring and API behaviour (Pytest).

Baseline comparison

RESULTS PENDING

Measure	Fixed roadmap	Score-only	Proposed
Skill coverage	TBD	TBD	TBD
Prerequisite compliance	TBD	TBD	TBD
Time-budget compliance	TBD	TBD	TBD
Explanation consistency	TBD	TBD	TBD

We will test with fictional learner profiles. This checks the roadmap logic; it does not prove real students learn better — that would need a separate, approved study.

PROJECT PLAN

Implementation Timeline and Milestones

Proposed 16-week work plan



2 Dec 2026
Mid-semester report

20 Jan 2027
Final report

Week 16
Final demonstration

CONCLUSION

Expected Outcomes and Limitations

What we expect to deliver — and what we will not claim

Expected outcomes

- A working, explainable career-preparation web app
- Evidence-based skill feedback for learners
- Prerequisite-aware, time-fitted roadmaps with reasons
- A measured comparison against two baselines

Limitations

- One career track in the first release
- Small, curated question bank and skill map
- Algorithm evaluation, not learning-outcome proof
- Does not predict employability or guarantee placement

CareerPilot AI supports structured learning decisions — assess fairly, explain every recommendation, and evaluate before claiming improvement.

THANK YOU

Thank You — Questions?

We welcome feedback from our supervisor and reviewers

KEY TAKEAWAYS

1. Recommendations are based on assessment evidence
2. Unknown evidence is never treated as zero skill
3. Prerequisites and weekly time shape the roadmap
4. Every recommendation is explained

PRESENTED BY

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SUPERVISOR

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CareerPilot AI is a proposed project. All examples, screens and values in this presentation are illustrative.